

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization With Tableau Practical – IV

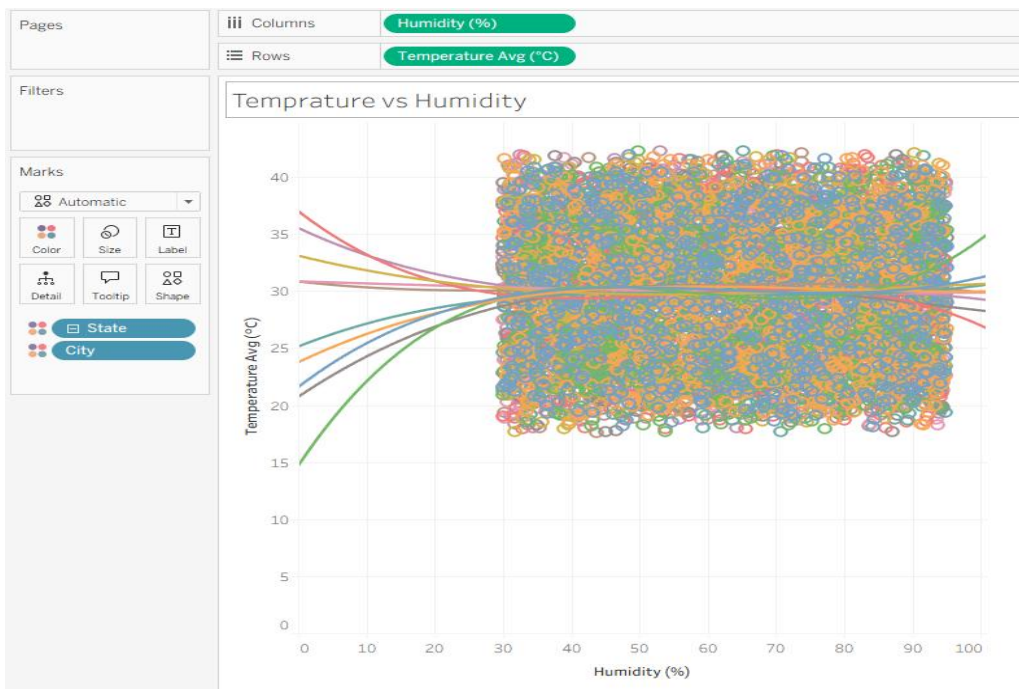
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Class :- TYIT	Batch :- 1
Date of Assignment :- 27/08/2026	Date/Time of Submission :- 27/08/2026

Performing Advanced Analytics using Trends, Clustering, and Forecasting

A) Add and interpret trend lines (linear, exponential)

Steps:

1. Open **Tableau Desktop** and load `Indian_Climate_Dataset_2024_2025.csv`.
2. Open a new worksheet and rename it to **Temperature vs Humidity Trends**.
3. Drag **Humidity (%)** to the **Columns** shelf.
4. Drag **Temperature_Avg (°C)** to the **Rows** shelf.
5. Go to **Analysis** and uncheck **Aggregate Measures** so that each data point is plotted individually.



B) Customize and analyze trend models

Steps:

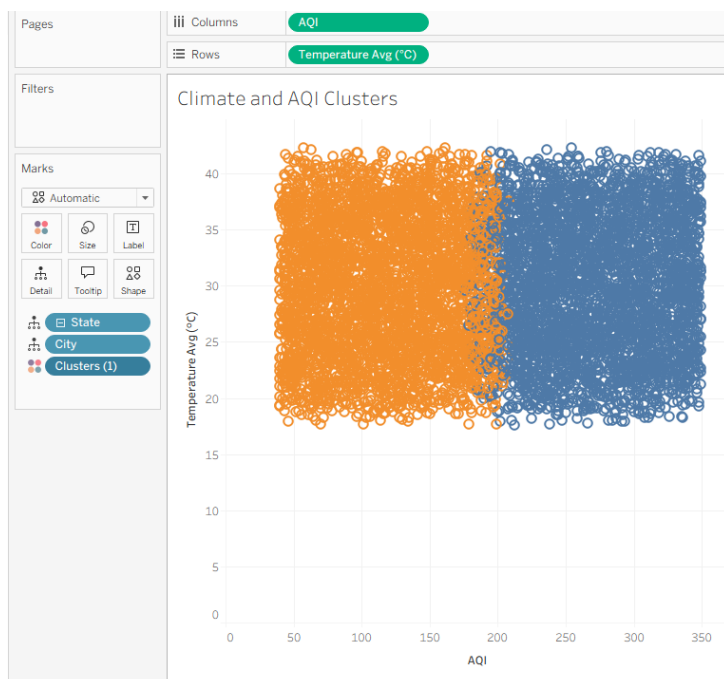
1. Open the **Temperature vs Humidity Trends** worksheet.
2. Drag **City** to **Color** on the Marks card.
3. Right-click the trend line and select **Edit Trend Lines**.
4. Change the model type to **Polynomial (Degree 2)** and click **OK**.
5. Use **Describe Trend Line** to check the equation, R^2 and P-value.



C) Perform clustering on datasets

Steps:

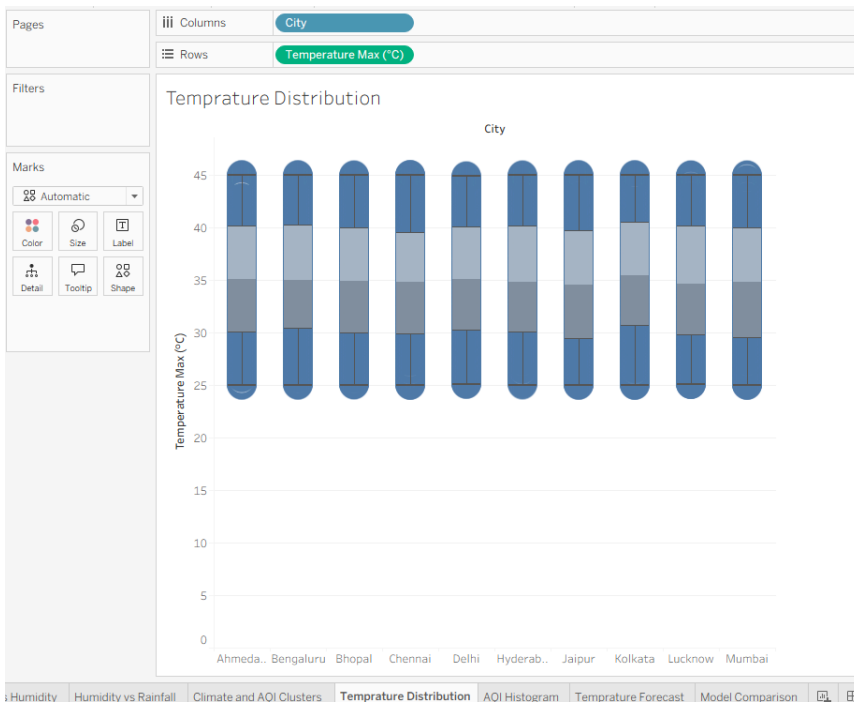
1. Create a new worksheet named **Climate & AQI Clusters**.
2. Drag **AQI** to Columns and **Temperature_Avg (°C)** to Rows.
3. Open the **Analytics** pane and drag **Cluster** onto the canvas.
4. Add **AQI, Temperature, Humidity and Wind Speed** as clustering variables.
5. Keep the number of clusters **Automatic** or select 3–4 clusters.



D) Analyze distributions using built-in tools

Steps:

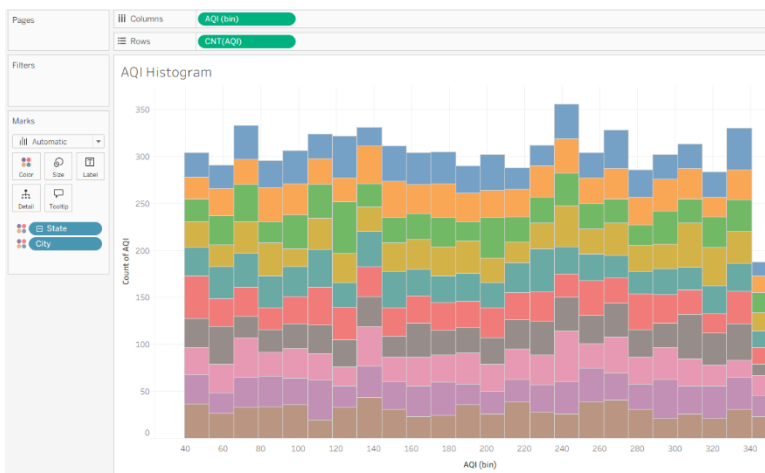
1. Create a new worksheet named **Temperature Distributions**.
2. Drag **City** to Columns and **Temperature_Max (°C)** to Rows.
3. Open the **Analytics** pane and drag **Box Plot** onto the chart.
4. Create another worksheet named **AQI Histogram** and select **AQI**.
5. Open **Show Me** and select the **Histogram** icon.



E) Apply forecasting techniques

Steps:

1. Create a new worksheet named **Temperature Forecast**.
2. Drag **Date** to Columns and change it to **continuous Month**.
3. Drag **Temperature_Max (°C)** to Rows and set it to **AVG**.
4. Open the **Analytics** pane and drag **Forecast** onto the chart.
5. Open **Forecast Options** and set the forecast length to **6 Months** with a **95% prediction interval**.

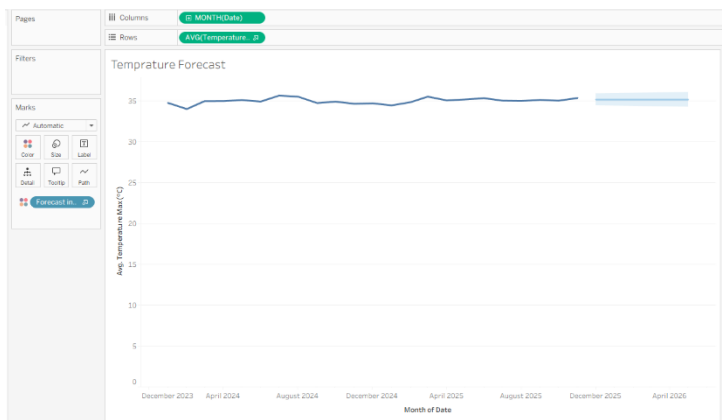


F) Compare different statistical models

Steps:

1. Create a new worksheet named **Model Comparison**.

2. Drag **Temperature_Max (°C)** to Columns and **AQI** to Rows.
3. Drag **City** to Detail on the Marks card.
4. Add a **Linear Trend Line** from the Analytics pane.
5. Edit the trend line and compare **Linear** and **Polynomial** models using their R^2 values.



G) Export and interpret statistical results

Steps:

1. Open any worksheet containing a trend line or forecast.
2. Right-click the trend line or chart.
3. Select **Describe Trend Model** or **Describe Forecast**.
4. Click **Copy** or **Export** to save the statistical results.
5. Check the **equation parameters**, **R^2 value** and **P-value** and record the findings.

